Lecture Scribe LaTeX

Run 1:
Reasoning:
From the lecture images, the lecture starts with probability basics → definitions → axioms → propositions → assigning probability → joint probability → conditional probability → examples (cards, dice, poker, key problem).
To generate the scribe, I will follow slide order and explain each concept step-by-step using reasoning before writing the final statement.

Scribe with reasoning

1. **Lecture Topic Identification**
   From first slide, topic is *Joint Probability and Conditional Probability*.
   So the lecture first reviews probability theory, then moves to joint and conditional probability.

2. **Basic Definitions reasoning**
   Slides define experiment, outcome, event, sample space.
   Reason: probability needs sample space before assigning probability.

Scribe:
Experiment = procedure producing result
Outcome = possible result
Event = set of outcomes
Sample space = all outcomes

3. **Axioms reasoning**
   To assign probability, axioms required.
   Slides show
   $0 \leqslant P(A) \leqslant 1$
   $P(S)=1$
   Additivity for disjoint events

Scribe:

Probability follows axioms
$P(A^c)=1-P(A)$
$P(A\cup B)=P(A)+P(B)-P(A\cap B)$

4. **Assigning probability reasoning**
   Slides show classical and relative frequency.
   Coin, dice, simulation table.

Scribe:
Classical → equally likely
Relative frequency → experiment repetition

5. **Joint probability reasoning**
   Slides show motivation: events not mutually exclusive.
   Notation $P(A,B)$.

Scribe:
Joint probability = probability of intersection.

Example card deck solved using counting.

6. **Conditional probability reasoning**
   Slides define
   $P(A|B)=P(A,B)/P(B)$

Then product rule and chain rule.

Scribe:
Conditional probability used when events depend.

Example cards without replacement, poker, missing key.

Conclusion:
Lecture covers full probability foundation → joint → conditional.

Run 2:
Reasoning:

Follow slide order again to keep same context.
Check consistency:
definitions → axioms → propositions → probability assignment → joint → conditional → examples.

Scribe with reasoning

1. Topic reasoning
   Slide title shows Lecture 4 probability.
   So lecture builds on previous lecture.

Scribe:
Lecture explains probability theory needed for joint and conditional probability.

2. Definitions reasoning
   Need experiment first → then outcome → event → sample space.

Scribe:
Experiment produces outcomes
Sample space contains all outcomes
Event is subset of sample space.

3. Axioms reasoning
   Probability must satisfy rules.

Scribe:
$0 \leqslant P(A) \leqslant 1$
$P(S)=1$
Mutually exclusive add.

4. Propositions reasoning
   Slides show complement rule and union rule.

Scribe:
$P(Ac)=1-P(A)$
$P(A \cup B)=P(A)+P(B)-P(A \cap B)$

5. Assign probability reasoning
   Two methods shown.

Scribe:
Classical counting
Relative frequency simulation

6. Joint probability reasoning
   Need intersection probability.

Scribe:
$P(A,B)=P(A\cap B)$
Use counting or frequency.

Card example confirms.

7. Conditional probability reasoning
   Definition slide.

Scribe:
$P(A|B)=P(A,B)/P(B)$

Product rule:
$P(A,B)=P(A|B)P(B)$

Chain rule for many events.

Examples confirm formula.

Conclusion reasoning
Lecture goal = understand dependence between events.

Run 3:
Reasoning:
Use same order again.
Check agreement with slides.
Ensure same structure as slides.

Scribe with reasoning

1. Topic reasoning
   Lecture title shows joint and conditional probability.

Scribe:
Lecture reviews probability then introduces joint and conditional probability.

2. Definition reasoning
Slides show experiment → outcome → event → sample space.

Scribe:
Sample space contains all possible outcomes
Events are subsets.

3. Axioms reasoning
Slides list three axioms.

Scribe:
Probability between 0 and 1
Whole space probability =1
Additivity rule.

4. Propositions reasoning
Slides derive formulas.

Scribe:
Complement rule
Union rule
General addition rule.

5. Assigning probability reasoning
Slides show coin, dice, simulation.

Scribe:
Classical method
Relative frequency method.

6. Joint probability reasoning
Slides define notation.

Scribe:
Joint probability = intersection probability
Use counting.

Examples: cards, clothes.

7. Conditional probability reasoning
   Slides define formula.

Scribe:
P(A|B)=P(A,B)/P(B)

Product rule and chain rule.

Examples:
cards without replacement
poker flush
missing key.

Conclusion reasoning
Lecture explains probability rules needed for dependent events.

Selected Output:
Run 2

Reason for selection:

* Same assumptions as other runs
* Same slide order
* Same formulas
* Same reasoning steps
* No missing concept
* Highest agreement across runs.

give me overlead latex code

Below is the **Overleaf LaTeX code** for your **Self-Consistency Scribe (Run 1, Run 2, Run 3 + Selected Output)**.
This code is clean, with spacing between runs, proper headings, and good format for assignment / scribe submission.

✅ Overleaf LaTeX Code

LaTeX

[redacted]

[redacted]

✅ This code is perfect for

- Assignment submission
- Lecture scribe
- Self-Consistency prompting task
- Overleaf PDF

✅ If you want, I can also make **more professional format (teacher impressed version).**